

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnauy W.		
Certified By	09-Jun-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-005-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-007**

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**

WPS No. **RNC-WPS-007**

Date **9-Jun-12**

Revision No. **0**

Supporting PQR No.(s) **RNC-PQR-007**

Date **9-Jun-12**

Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding (GTAW)**

(Automatic, **Manual**, Machine, or Semi-Auto.)

JOINT (QW-402)

Joint Design **Single V**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **8** Group No. **1** to P-No. **8** Group No. **1**

Material Specification Type and Grade **A 312 Gr.TP 316L, A182 F316L** TO **A 312 Gr.TP 316L, A182 F316L**

Chem. Analysis and Mech. Prop.

Thickness Range :

Base Metal : Groove **1.6 mm - 7.82 mm.** Fillet **All**

Pipe Dia. Range : Groove **All** Fillet **All**

Other

FILLER METALS (QW-404)

GTAW

Spec. No. (SFA) **5.9**

AWS No. (Class) **ER-316L**

F-No. **6**

A-No. **8**

Size of Filler Metals **Ø 2.0 & 2.4 mm**

Deposited Weld Metals **3.91 mm.**

Thickness Range :

Groove **7.82 mm Max.**

Fillet **All**

Electrode-Flux (Class) **-**

Flux Trade Name **-**

Consumable Insert **-**

Other **Kobelco TGS-316L or Equivalents**

POSITION (QW-405)

Position(s) of Groove **ALL**

Welding Progression Up ☒ X Down ☐

Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **N/A**

Time Range **N/A**

Notes

PREHEAT (QW-406)

Preheat Temp. Min. **10°C**

Interpass Temp. Max. **150°C**

Preheat Maintenance **N/A**

Notes

GAS (QW-408)

Percent Composition

	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	12-16 Ltr/Min
Trailing	-	-	-
Backing	Argon	99.99%	13-17 Ltr/Min

WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WPS-007 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☐ AC ☒ DC Polarity EN
 Amps (Range) 70-130 Volts (Range) 8-13
 Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

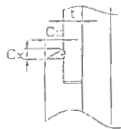
String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) 5-11 Cm/Minute
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER-316L	2.4	DC EN	80-110	8-12	5-9	
2nd	GTAW	ER-316L	2.4	DC EN	100-130	10-13	7-11	
Other(s)	GTAW	ER-316L	2.4	DC EN	70-120	10-13	6-9	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of **ASME IX, 2007 latest Revision**.

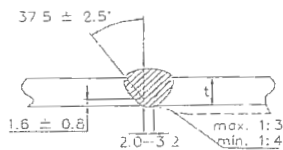
Prepared by: Mr. Chatchanan S.  9/06/12
 Name Signed Date
 Reviewed by: Mr. Sansan ch 9-06-12
 Name Signed Date
 Approved by: Amnuay W. 29-06-12
 Name Signed Date

ATTACHED FIGURE JOINT SKETCH



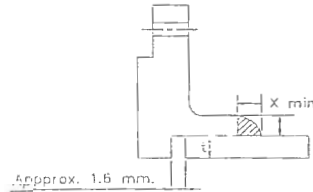
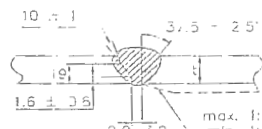
C_x (min) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



Misalignment (max.) = 1.5 mm.

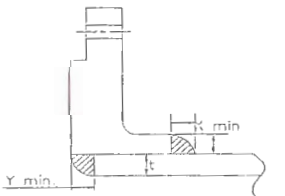
JOINT No. B1 & B2



Approx. 1.6 mm.

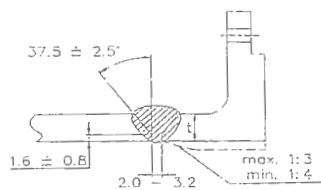
X min = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



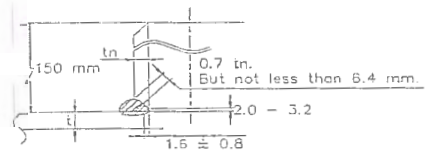
X min = The lesser of $1.4t$ or thickness of the hub.
 Y min = The lesser of t or 6.4 mm.

JOINT No. A3



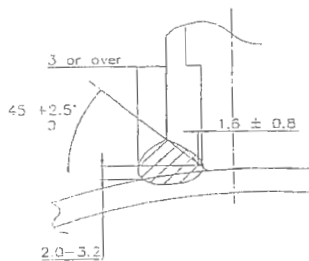
Misalignment (max.) = 1.5 mm.

JOINT No. B3

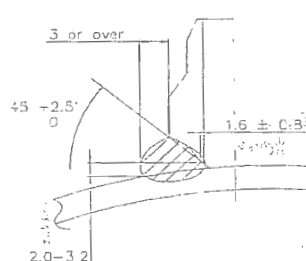


Misalignment (max.) = The lesser of 3.2 mm or 0.5tn.

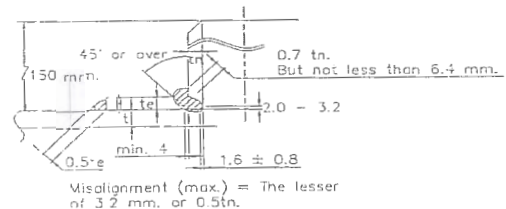
JOINT No. B4



JOINT No. C1

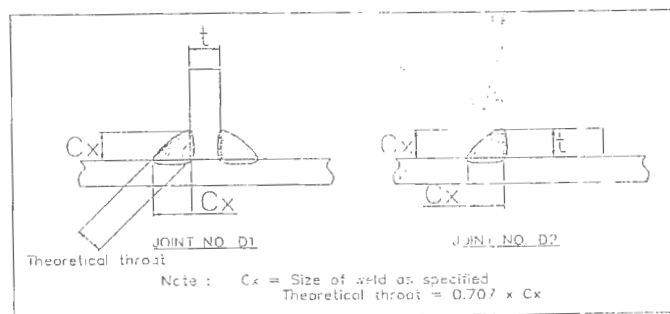


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm or 0.5tn.

JOINT No. B5



WELDING PROCEDURE

QUALIFICATION RECORDS

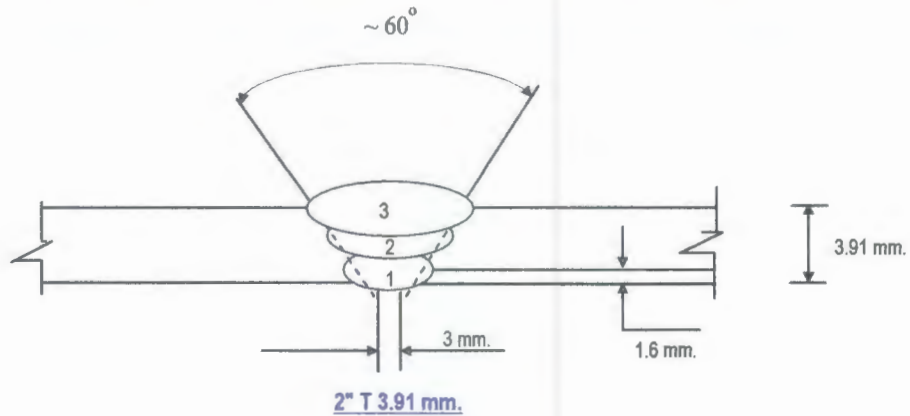
PQR No. : RNC-PQR-007

WPS No. : RNC-WPS-007

THIS PQR RECORDS ACTUAL TEST

CONDITION AND RESULT

Welding Sequences for PQR-RNC-007 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Wire Feed		Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER316L	2.4	55	10	DCEN	-	-	6	8.40	15	16
2	GTAW	ER316L	2.4	60	10	DCEN	-	-	9	7.06	15	16
3	GTAW	ER316L	2.4	60	10	DCEN	-	-	8	6.52	15	16

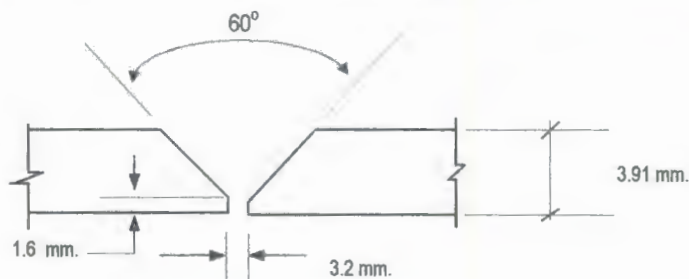
Procedure Qualification Record (PQR)

ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : **RNC-PQR - 007**

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-007	Date :	13 June 2012
WPS Number :	RNC-WPS-007		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)			
Material Spec.	A 312	To	A312	Temperature	N/A		
Type or Grade	TP316L	To	TP316L	Hold Time	N/A		
P. No.	P. 8	To	P. 8	Others	N/A		
Thickness of Test Coupon	3.91 mm.			GAS (QW-408)			
Diameter of Test Coupon	Ø 2"						
Others							
FILLER METALS (QW-404)				Percent Composition			
SFA Specification	A 5.9				Gas (es)	Mixture	Flow Rate
AWS Classification	ER316L			Shielding	Argon	99.95%	15 L/Min.
Filler Metal F-No.	6			Trailing	-	-	-
Weld Metal Analysis A-No.	8			Backing	Argon	99.95%	16 L/Min.
Size of Filler Metal	Ø 2.4 mm			ELECTRICAL CHARACTERISTICS (QW-409)			
Others	Kobelco (TGS-316L)			Current	DC		
Weld Metal Thickness	-			Polarity	EN		
POSITION (QW-405)				Amperage	84 - 106	Voltage	10
Position of Groove	6G			Tungsten Electrode Size	Ø 2.4 mm. 2% thoriated.EWTh-2		
Progression of Welding (Uphill or Down)	Up Hill			Others	N/A		
Others	N/A			TECHNIQUE (QW-410)			
PREHEAT (QW-406)				Travel Speed	6 - 9 cm/min.		
Preheat Temperature	Ambient Temperature			String or Weave Bead	Both		
Interpass Temperature (max.)	144° C			Oscillation	N/A		
Others	N/A			Multipass or single Pass	Multiple Pass		
				Multipass or Single Electrodes	Single Electrode		
				Others	N/A		

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- **PQR-RNC-007**

Tensile Test Report No.TS-12-06-107

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
2368/12 TR1	12.43	4.06	50.466	28.753	569.754	WELD ZONE
2368/12 TR2	13.12	4.08	53.530	30.165	563.524	WELD ZONE

Guided Bend Tests (QW-160) Report No.BD-12-06-105

Type and Figure Number	Type of Bend	Indication	Results
2368/12 BF1	Transverse face bend	No Open Discontinuity	Accepted
2368/12 BF2	Transverse face bend	No Open Discontinuity	Accepted
2368/12 BR1	Transverse root bend	No Open Discontinuity	Accepted
2368/12 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR -003
Hardness Testing Report No. HN-12-06-082
Macrostructure Report No. MA-12-06-104
Impact Testing Report No.IM-12-06-140

Welders Name Mr.Boontham J. Welder Number -
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :



Inspected by :

Certified by :

Date :

Mr.Nathawut M.

Mr.Chalermkiet R.

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WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1[illegible]

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME : _____

DATE : _____



SIGNATURE :

NAME :

DATE : _____

Witness By



SIGNATURE :

NAME :

DATE : _____

AMULY u
29-06-12

29-06-12



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Welding Procedure Specification (WPS)



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

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Procedure Qualification Record (PQR)



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Radiographic Examination Report

APPROVED BY

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)607402-3, Fax : 038)607402. SRIRACHA : 038)330220, 330229

Page : 1 Of 1[illegible]



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Mechanical Test Report



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210 Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnplm.co.th



22688

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1
Report No. TS-12-06-107
Revision No. 0

Test Product	PQR No. RNC-PQR-007	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A312 TP316L	Type of Test	Reduced Section Tension
Dimension	Ø 2" Sch 40 (3.91 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	19 June 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	21 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	2368/12 TR1	N/A	4.06	12.43	50.466	N/A	N/A	N/A	28.753	569.754	N/A	WELD ZONE
2	2368/12 TR2	N/A	4.08	13.12	53.530	N/A	N/A	N/A	30.165	563.524	N/A	WELD ZONE

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-007

Welding Process : GTAW

Filler Metal/Electrode Spec't : ER-316L/TGS-316L

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 25 / 07 / 12



Approved by:

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 25 / 07 / 12

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

22689

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmpm.co.th

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-12-06-105

Revision No. 0

Test Product	PQR No. RNC-PQR-007	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A312 TP316L	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 2" Sch 40 (3.91 mm Thk.)	Diameter of Mandrel	16 mm
Date Received	19 June 2012	Bend Angle	180 Degree.
Date Tested	21 June 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	2368/12 BF1	Transverse face bend	3.89x19.05x157.47 mm	No Open Discontinuity	N/A	
2	2368/12 BF2	Transverse face bend	4.03x19.04x160.14 mm	No Open Discontinuity	N/A	
3	2368/12 BR1	Transverse root bend	4.10x19.11x159.87 mm	No Open Discontinuity	N/A	
4	2368/12 BR2	Transverse root bend	3.94x19.05x159.34 mm	No Open Discontinuity	N/A	

Additional details : WPS No. RNC-WPS-007

Welding Process : GTAW

Filler Metal/Electrode Spec't : ER-316L/TGS-316L

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date05/07/12



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date05/07/12

LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th



22948

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1
Report No. IM-12-06-140
Revision No. 0

Test Product	PQR No. RNC-PQR-007	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ¹
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A312 TP316L	Type of Test	Charpy V-Notch Impact
Dimension	Ø 2" Sch 40 (3.91 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	19 June 2012	Equipment 's Capacity	542 Joules
Date Tested	22 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	2368/12-1	-20°C	2.519	10.014	54.72	33.82	32.18	Base Metal	-
2	2368/12-2	-20°C	2.515	10.011	54.71	30.35			
3	2368/12-3	-20°C	2.518	10.013	54.72	32.38			
4	2368/12-1	-20°C	2.504	10.014	54.83	32.17	31.42	Weld Metal	-
5	2368/12-2	-20°C	2.504	10.001	54.86	32.37			
6	2368/12-3	-20°C	2.504	10.001	54.82	29.73			
7	2368/12-1	-20°C	2.501	9.999	54.83	36.17	37.44	HAZ	-
8	2368/12-2	-20°C	2.504	10.000	54.87	37.92			
9	2368/12-3	-20°C	2.500	10.004	54.95	38.24			

Additional details : WPS No. RNC-WPS-007

Welding Process : GTAW

Filler Metal/Electrode Spec't : ER-316L/TGS-316L

Reported by :
(Mr.Keerati Sa-ngoen)
Laboratory Section Manager
Date 13 / 07 / 12

Approved by :
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date 13 / 07 / 12

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LPN PLATE MILL



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199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

11111

Macrostructure Analysis Report

Page 1 of 1

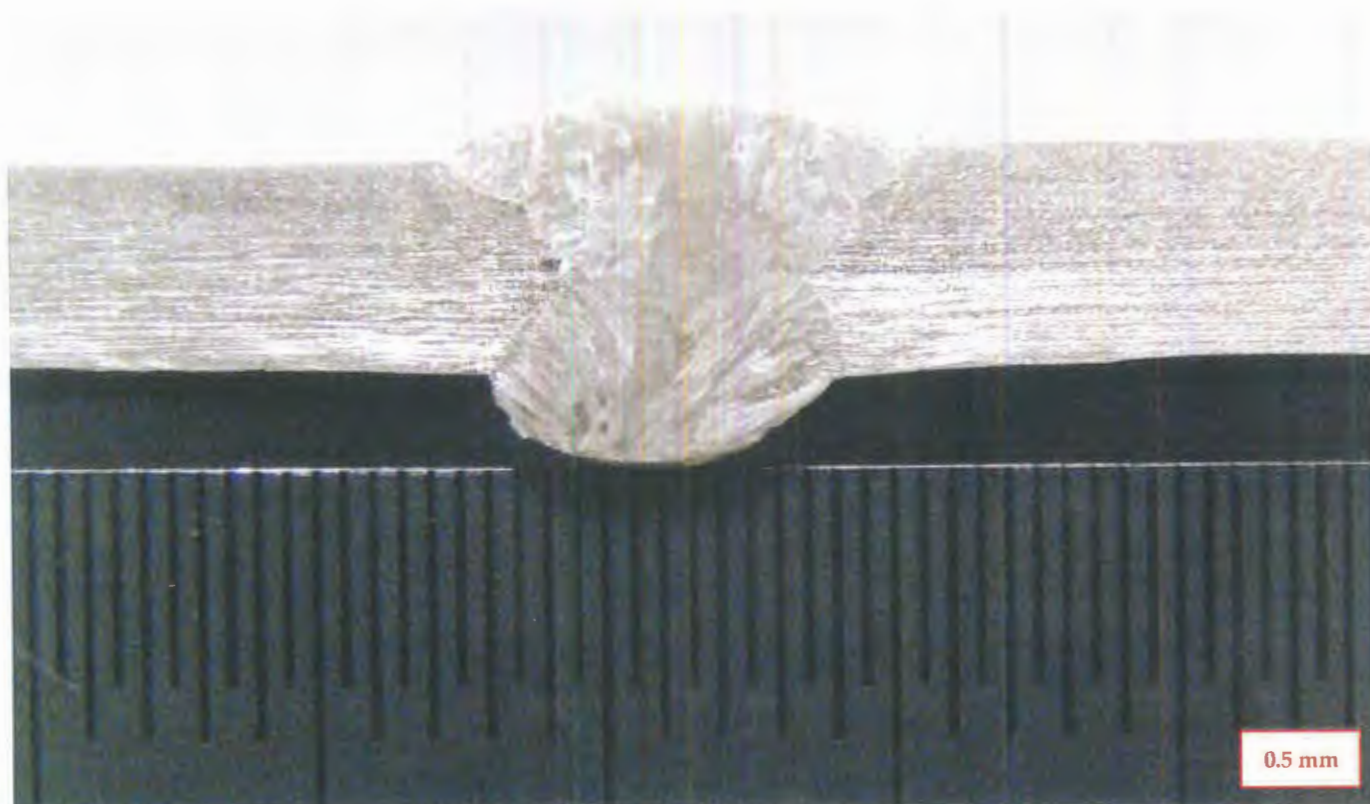
Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. MA-12-06-104

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-007	Material Characterization	Pipe
Test No.	2368/12 MA	Test Method	ASME IX : 2010
Project Name	N/A	Standard of Test	ASME IX : 2010
Project No.	N/A	Etchant Reagent	75ml H ₂ O+25ml HNO ₃ (65%)
Material Specification	A312TP316L	Temperature	28 °C
Dimension	Ø2"Sch40(3.91 mm Thk.)	Time of Etching	5 minutes
Welding Process/Position	GTAW	Surface Preparation	Polishing with sand paper
Type of Joint	Butt Joint		240, 400, 600, 800, 1000
Date Received	19 June 2012	Date Tested	21 June 2012



Additional details : The macrostructure shows the complete fusion.Weld metal and heat affected zone were free of cracks

Filler/Electrode Spec't : ER-316L TGS-316L

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date : 27.6.12



Approved by:

(Mr.Pavaret Freedawiphat)

Deputy Managing Director

Date : 27.6.12



Notes

1. The above results are valid exclusively for tested samples as mentioned in this report.
2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
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Testing Laboratory LPN Plate Mill Public Company Limited

12470

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnmpm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. HN-12-06-082

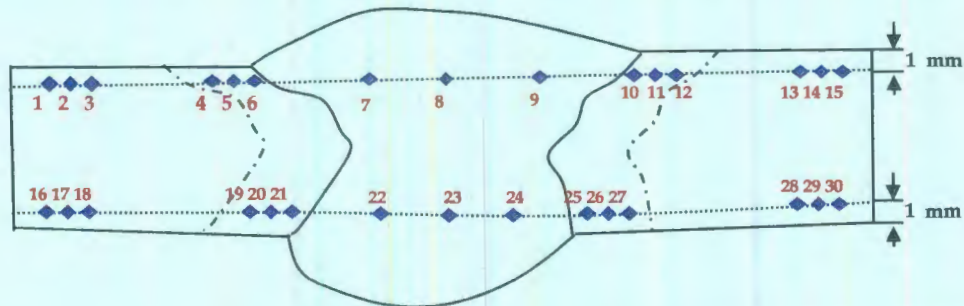
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-007	Material Characterization	Pipe
Test No.	2368/12 HN	Test Method	ASTM E92-82 : 2003
Project Name	N/A	Standard of Test	ASTM Sec. IX : 2010
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A312TP316L	Equipment	Mitutoyo, HV-115
Dimension	Ø2"Sch40(3.91 mm Thk.)	Indenter Size	Diamond, pyramid 136 deg.
Welding Process/Position	GTAW	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	19 June 2012	Cal. Block	No. 62137 No. A62072
Date Tested	21 June 2012	Ambient Temperature	25.8 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	173.9	16	BASE	173.0	*****		
2	BASE	173.6	17	BASE	172.8			
3	BASE	174.2	18	BASE	172.3			
4	HAZ	169.2	19	HAZ	179.8			
5	HAZ	163.5	20	HAZ	173.6			
6	HAZ	159.3	21	HAZ	160.5			
7	WELD	149.7	22	WELD	146.7			
8	WELD	154.1	23	WELD	154.8			
9	WELD	152.1	24	WELD	149.4			
10	HAZ	156.2	25	HAZ	160.7			
11	HAZ	154.7	26	HAZ	167.7			
12	HAZ	160.3	27	HAZ	158.3			
13	BASE	165.4	28	BASE	159.5			
14	BASE	166.1	29	BASE	160.1			
15	BASE	166.0	30	BASE	161.3			

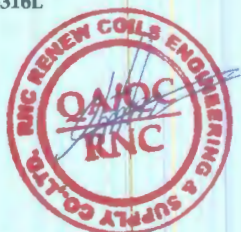
Additional details : Filler/Electrode Spec't : ER-316L TGS-316L

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date : 23/07/12

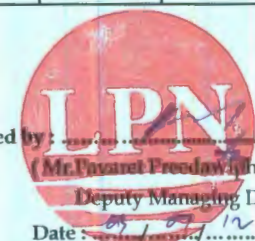


Approved by :

(Mr.Pavanet Freedawnghat)

Deputy Managing Director

Date : 23/07/12



LPN PLATE MILL

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Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 -9978

Consumable Certificate

TKW120106-KOBELCO

INSPECTION CERTIFICATE
(検 査 証 明 書)PURCHASER
需要家WELDING ROD OR WIRE
溶加棒又はワイヤCERTIFICATE No
証明書番号 : 220120244
DATE OF ISSUE
発行日 : 2012.03.21

TRADE DESIGNATION 品 名 (型 番)	DIMENSION 寸 法 (mm)	MFG.No 製 造 番 号	APPLICABLE SPECIFICATION AND CLASSIFICATION 適 用 規 格 及 び 種 類
TG-S316L	(2.4) mm 3/32 inch	E36762	AWS A5.9 ER316L ASME SFA-5.9 ER316L

CHEMICAL COMPOSITION 化学成分(%) ACCORDING TO EN10204 TYPE 3.1

ELEMENTS 成分	C	SI	MN	P	S	CU	NI	CR	MO	NB	N	FN	FS
ROD OR WIRE 棒又はワイヤ	0.008	0.40	1.55	0.024	0.002	0.15	12.18	18.58	2.14	<0.01	0.02	10	6
ELEMENTS 成分	FNW							FS = FERRITE % (SCHAEFFLER DIAGRAM)					
ROD OR WIRE 棒又はワイヤ	9							FN = FERRITE NUMBER (DELONG DIAGRAM)					
								FNW= FERRITE NUMBER (WRC)					

TENSILE PROPERTIES OF DEPOSITED METAL 溶着金属の引張性能				IMPACT PROPERTIES OF DEPOSITED METAL 溶着金属の衝撃性能				HARDNESS PROPERTIES 硬さ
YIELD POINT 降 伏 点	YIELD STRENGTH AT 0.2% OFFSET 0.2% 耐力	TENSILE STRENGTH 引 張 強 さ	ELONGATION 伸 び	TEST TEMP. 試験温度	ABSORBED ENERGY 吸 収 エ ネ ル ギ ー			
(-----) N/mm ² ----- ksi	(-) N/mm ² - ksi	(-----) N/mm ² ----- ksi	- %	(-) °C - °F	AVG. 平均 (-) J - ft-lbf	EACH ()		-

WELDING CONDITIONS 溶 接 条 件				POSTWELD HEAT TREATMENT 溶 接 後 熱 処 理	
TYPE OF CURRENT 電 流 の 種 類	-	SHIELDING GAS シールドガス	-		
AMPERAGE 溶 接 電 流	- A			(-) °C × - h	
ARC VOLTAGE アーク電圧	- V			- °F × - h	

WE HEREBY CERTIFY THAT THE ABOVE TEST RESULTS
AND MECHANICAL PROPERTIES SPECIFICATIONS SATISFY
THE REQUIREMENTS OF THE APPLICABLE SPECIFICATION.上記試験結果および機械的性質保証値は、適用規格の要求を
満足していることを証明します。

◆KOBELCO STEEL, LTD.

WELDING BUSINESS FUJISAWA WORKS
株式会社 神戸製鋼所 溶接事業部門 藤沢事業所

N. Cerachi

APPROVED BY CHIEF INSPECTOR
主任検査員 承認済



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 -9978

Material Certificate

CERTIFIED MATERIAL TEST REPORT

E3

Messrs. UR MENA LTD	Date Apr. 20, 2011
Note Grade TP 316/316L	Sheet No. 627556
Condition Solution Treated	Contract No. T13TH26
Heat No. 801971	Packing List No. ET3827
Size 2 X SCH 40.S	Bundle No. 3 - 4
Length 6,000 - 6,003 mm	
Pieces 35	
Net Weight 1,143 kg	

SANYO SPECIAL STEEL CO., LTD.

Quality Assurance Department
Quality Certification Group
3007, Nakashima, Shikama-ku, Himeji, Japan
TEL (079) 235-6029
FAX (079) 235-6227

Chemical Composition (%)	C	Si	Mn	P	S	Ni	Cr	Mo
	X1000	X100	X100	X1000	X1000	X100	X100	X100
Standard	35	100	200	45	30	1100	1600	200
	max	max	max	max	max	1400	1800	300
Cast	14	46	127	36	4	1123	1626	206

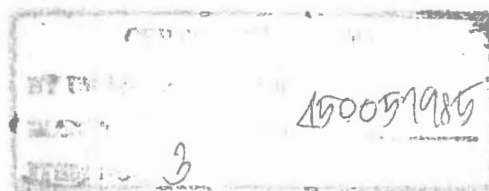
Mechanical Test	Type of Specimen	0.2% Yield Strength	Tensile Strength	Elongation
From Standard Product		30 min Ksi	75 min Ksi	35 min %
Result	ASTM	43	80	50

Corrosion Test	(ASTM A262-E) : GOOD
Flattening Test	GOOD
Dimension Inspection	GOOD
Spark Test	GOOD
Eddy Current Test	GOOD
Specification	ASTM A312 ASME SA312
Remarks	Heat Treatment : 1070°C X 13M W.C SEAMLESS STAINLESS STEEL PIPES HARDNESS ACC. TO NACE MR0175:GOOD PMI is based on Spark Test DIN3.1 according to DIN EN 10204 (DIN 50049)



SS 316
CR. 16.26
NI. 11.23
MN. 1.27
MO. 2.06

payser
30/11/11



We hereby certify that the material described herein has been made in accordance with the rules of the contract.

Head of Quality Certification Group

K. Hashizume